Luke Wortsmann

Quantitative Developer

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Experience

Current
May 2024

Aquatic Capital Management | Quantitative Developer

Aquatic is a quantitative hedge fund, running a mid/high frequency stat-arb strategy.

- responsible for core components of the research platform and trading systems as part of the monetization group.
- focused on alignment between simulation and live trading.
- built and maintained code for portfolio optimization, alpha attribution, market impact modeling, execution algo A/B testing, and transaction cost analysis.
- collaborated with core engineering and research teams to deploy large-scale machine learning models and real-time alpha computation systems.
- managed code releases and built tools to monitor, test, and safeguard production trading environments.

2021 – 2024

OTT Risk | Senior Data Scientist

OTT was a venture-backed startup using machine learning to offer parametric business interruption insurance.

- structured parametric insurance programs, developed indices to facilitate transfer of novel risk.
- designed and implemented pricing and risk models in Stan and PyTorch, presented and defended models to reinsurers and other partners.
- managed data processing and model training pipelines, APIs (FastAPI), and dashboards (Dash, Plotly) deployed on Google Cloud Platform.
- served as primary software engineer for a seven-person startup; contributed to product development, client and vendor relations, fundraising, and regulatory compliance.

2017 – 2021

SBB Research Group | Quantitative Researcher

SBB is a quantitative hedge fund, primarily trading structured products and OTC derivatives.

- member of a small team responsible for research, risk management, and trading of a diverse portfolio of structured products, equities, options, and OTC derivatives.
- built pricing models for bespoke derivatives including those linked to equity indices, commodities, credit risk, foreign exchange rates, volatility, and correlation.
- implemented valuation models and risk management tools, from the ground up, with Python and C.
- developed proprietary market signals from fundamental and alternative datasets using TensorFlow, SQL, and R.
- coordinated implementation of trading strategies and internal tooling with software engineers and stakeholders across the firm.
- managed and mentored interns on the quantitative research team.

Summer 2016

KCG Holdings | Quantitative Trading Intern

KCG was a global, high frequency, trading firm and market maker.

- researched proprietary trading signals using machine learning and Bayesian methods.
- analyzed large, complex market datasets using Python, Vertica, and scikit-learn.
- collaborated with traders and other interns on the US equities desk.

Education

2014 – 2017

University of Illinois Urbana-Champaign | College of Engineering

Bachelor of Science **Physics** | Minor **Mathematics**

3.71 / 4.00 GPA. *degree completed in three years.*

Course work in: Numerical PDEs, Differential Geometry, and Probability Theory

Publications

March 2018

Large energy density in three-plate nanocapacitors due to Coulomb blockade.

Journal of Applied Physics *with Alfred Hubler, S. Foreman, and J. Liu.*